IR Transmission Research for Multi-Transmitter Systems

Abstract

This research investigates infrared (IR) transmission challenges in multi-transmitter systems, focusing on interference mitigation and system expansion. Initial experiments revealed interference between closely placed transmitters using the NEC protocol. Various isolation methods, including physical barriers and frequency-division multiplexing, are explored. A high-power IR LED driver circuit is proposed to overcome limitations of standard modules, enabling reliable communication across a 12×12 receiver array spanning $900\,\mathrm{cm}^2$.

1 Introduction

While developing the NEC parallel transmitter, interference was observed when two transmitters were placed in close proximity. Therefore, effective protection against this interference—such as guiding tubes or other isolation methods—must be developed.

2 Signal Characteristics

The signal adheres to the NEC standard with a carrier frequency of **38 kHz**. The transmission LED operates at a wavelength of **940 nm** [3], placing it in the **NIR (Near-Infrared)** region.

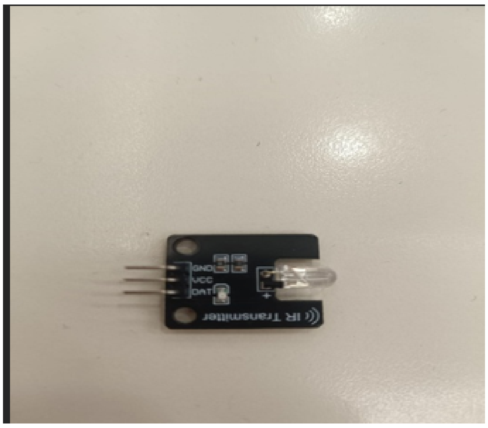


Figure 1: IR Transmitter Element

The spectral regions are illustrated in Figure 2.

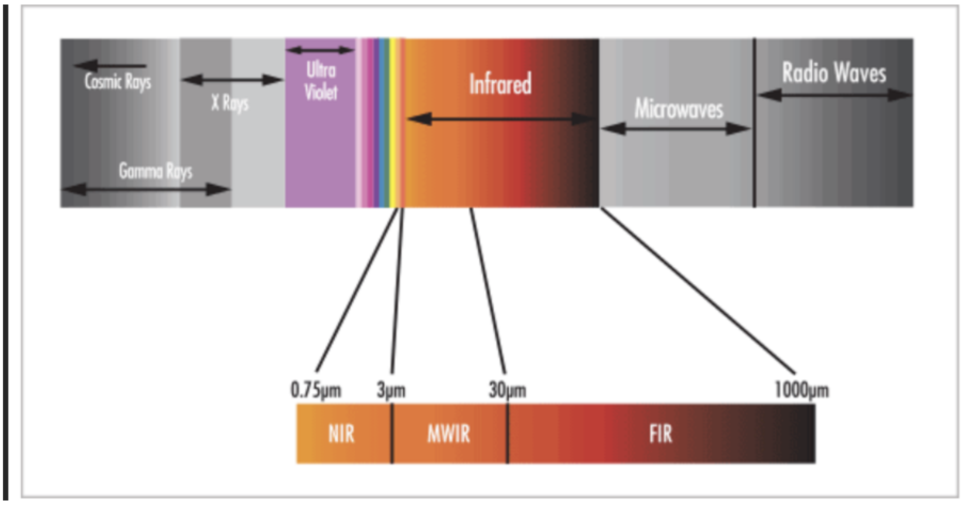


Figure 2: Infrared Spectral Regions [2]

3 System Expansion

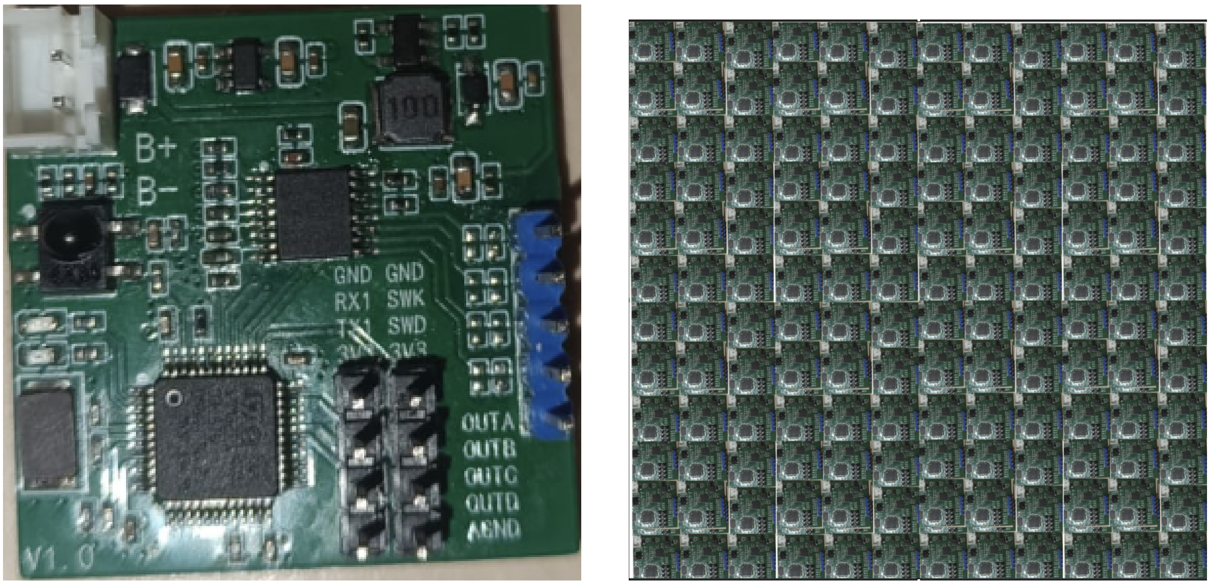
Additional parallel transmitters were required. The maximum number of available pins for transmission is:

- 8 pins from PMOD A
- 8 pins from PMOD B
- 20 pins from Arduino

This yields a total of **36 pins** for transmission. The final configuration uses 24 transmitters: all Arduino pins plus 4 PMOD A pins.

4 Single-Transmitter, 144-Receiver Array Configuration

The experimental setup comprises a single infrared transmitter paired with an array of 144 individual IR receivers. Each receiver unit occupies a footprint of $2.5\,\text{cm} \times 2.5\,\text{cm}$. The receivers are arranged in a contiguous 12×12 grid, forming a square detection surface with a total active area of $900\,\text{cm}^2$.



(a) Single IR receiver unit (2.5 × 2.5 cm) (b) Complete 12 × 12 receiver grid (900 cm^2)

Figure 3: Infrared receiver configuration

4.1 Problems with the IR Transmission

The initial prototype used the KY-005 infrared transmitter module [3]. However, this module proved inadequate for the required application due to its limited optical output power. The KY-005 is fundamentally a low-power, narrow-beam IR transmitter consisting solely of a 5 mm IR LED [5]. Its key specifications are:

Table 1: KY-005 Module Specifications

Parameter	**Value**
Operating Voltage	5 V
Forward Current	30–60 mA
Power Consumption	90 mW
Operating Temperature	−25°C to 80°C
Dimensions	18.5 × 15 mm
Compatibility	Arduino, Teensy, Raspberry Pi, ESP8266

Preliminary testing confirmed that the KY-005 module cannot reliably illuminate the entire 12×12 receiver grid. The module's restricted radiant intensity and narrow emission cone cause significant signal attenuation at the grid periphery, rendering it unsuitable for wide-area coverage.

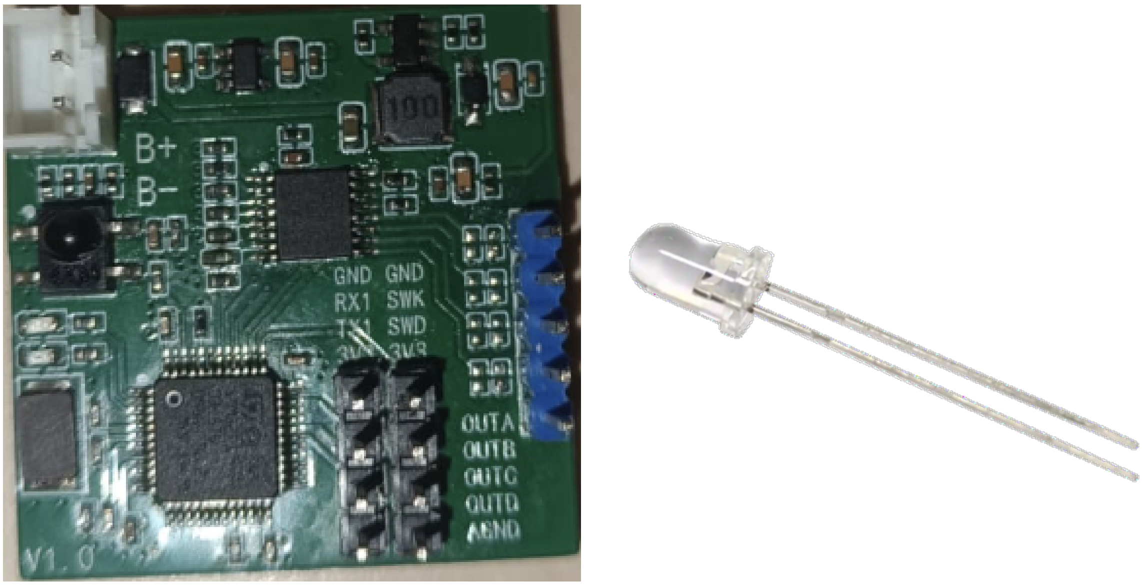
4.1.1 Proposed Solution: High-Power IR LED Array

To overcome these limitations, the KY-005 module will be replaced with a discrete high-power infrared LED driven by a dedicated external circuit. The selected IR LED (Lite-On [6]) offers substantially enhanced performance:

Table 2: High-Power IR LED Specifications

Parameter	**Range**
Wavelength	850–940 nm
Beam Angle	16° to 150°
Forward Voltage (V_F)	0.8–2.5 V
Reverse Voltage (V_R)	4–40 V
Forward Current (I_F)	1 mA to 1 A (pulsed)
Power Rating	100 mW to 2.5 W
Radiant Intensity	2.165–400 mW/sr
Rise/Fall Time	7.5–40 ns
Peak Soldering Temperature	+260°C

The increased forward current capacity (up to 1 A pulsed) and higher radiant intensity ensure sufficient optical flux to saturate all 144 receivers across the $900\,\mathrm{cm}^2$ area. The driver circuit modulates the LED at the required 38 kHz carrier frequency while supplying necessary peak current without exceeding thermal limits.



(a) IR receiver unit

(b) High-power IR transmitter

Figure 4: Enhanced IR components

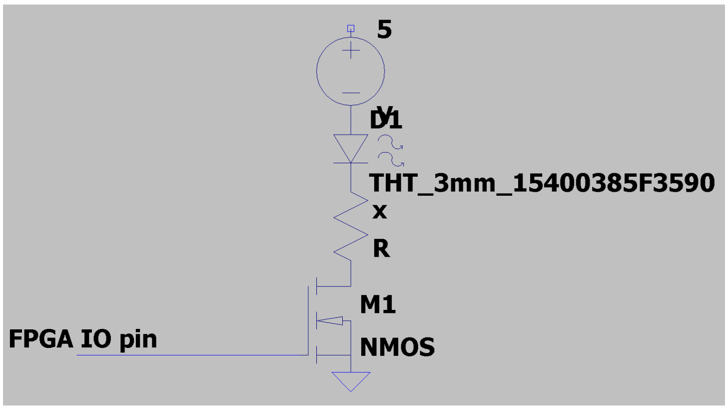


Figure 5: High-current driver circuit schematic for the high-power IR LED

4.1.2 Parameter Selection

The PYNQ Z2 FPGA cannot provide sufficient current to drive Lite-On IR LEDs directly. Therefore, a MOSFET driver circuit is required.

4.2 High-Power Solution (1A Operation)

For applications requiring 1A of drive current, a more robust circuit is necessary to protect both the FPGA and the IR LED. Figure 6 shows the complete schematic for driving a 1A IR LED with 38kHz NEC modulation.

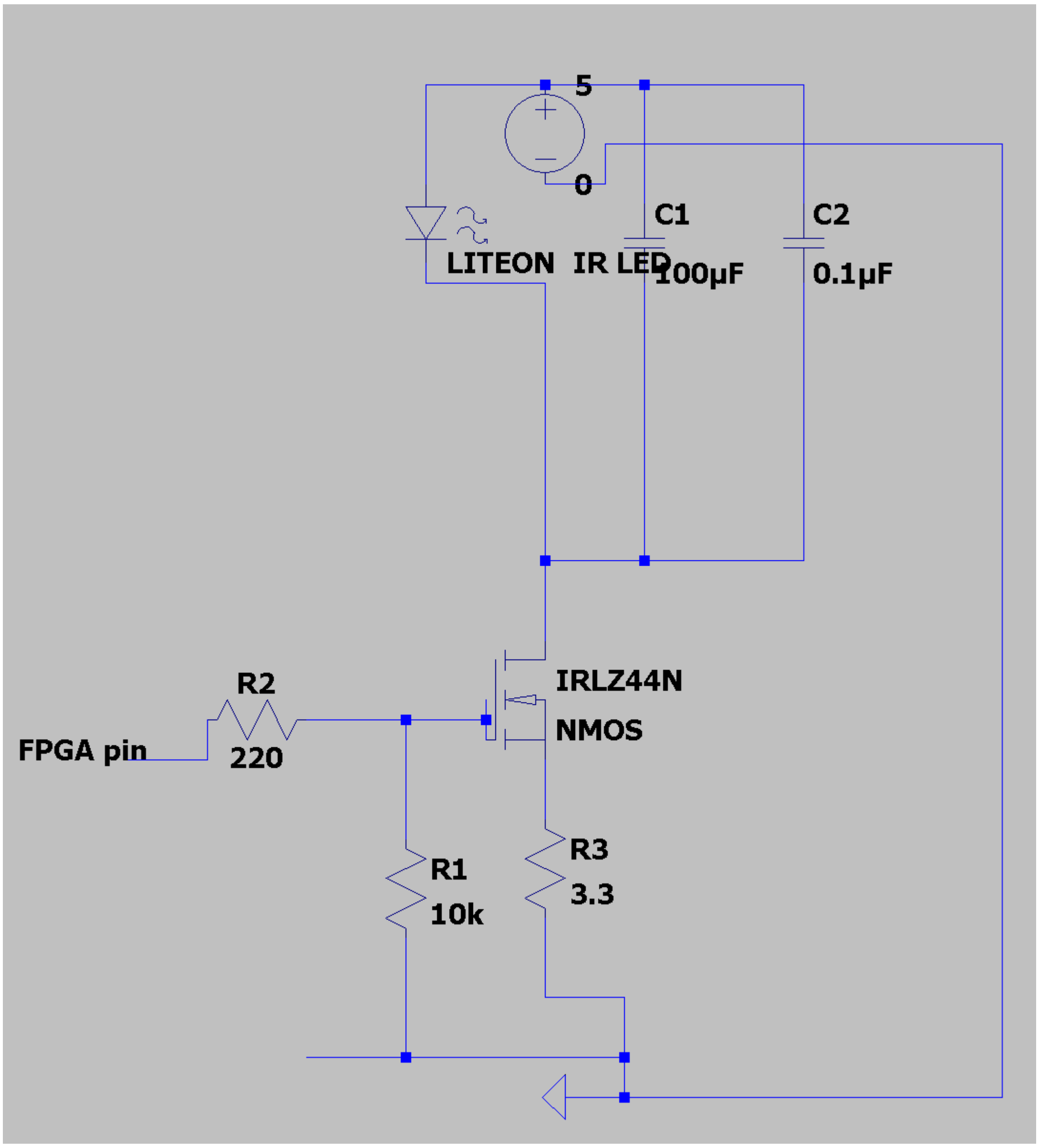


Figure 6: Complete 1A IR LED driver circuit with MOSFET switching.

4.2.1 Circuit Specifications

The circuit shown in Figure 6 implements a simple and reliable MOSFET switch configuration with the following specifications:

Table 3: Circuit Component Specifications

Component	**Value / Part Number**	**Purpose**
IR LED	High-power, 940nm, 1A capable	Emits infrared light
MOSFET (Q1)	IRLZ44NPbF	Logic-level N-channel power switch
R1 (Gate pull-down)	10kΩ, 1/4W	Ensures MOSFET is OFF during FPGA boot
R2 (Gate resistor)	220Ω, 1/4W	Limits current from FPGA pin, protects output
R3 (Current limit)	3.3Ω, 5W	Limits LED current to approximately 1A
C1 (Electrolytic)	100μF, 16V	Provides local charge storage for current spikes
C2 (Ceramic)	0.1μF, 50V	Filters high-frequency noise
Power Supply	5V DC, 2A minimum	Powers the entire circuit
FPGA Input	3.3V logic, 38kHz PWM	NEC protocol modulation signal

4.2.2 Circuit Calculations

The resistor values are calculated using Ohm's Law:

Current-Limiting Resistor (R3):

$$V_{\text{supply}} = 5\ \text{V}, \quad V_{\text{LED}} \approx 1.5\ \text{V}, \quad V_{\text{DS}} \approx 0.1\ \text{V}$$

$$V_R = 5 - 1.5 - 0.1 = 3.4\ \text{V}$$

$$R_3 = \frac{V_R}{I} = \frac{3.4}{1.0} = 3.4\ \Omega \quad (\text{use } 3.3\Omega)$$

$$P_{R3} = I^2 \times R = (1.0)^2 \times 3.3 = 3.3\ \text{W} \quad (\text{use 5W resistor})$$

Gate Resistor (R2):

$$R_2 = 220\ \Omega \quad (\text{limits FPGA pin current to 15mA})$$

$$I_{\text{FPGA}} = \frac{3.3\ \text{V}}{220\ \Omega} \approx 15\ \text{mA} \quad (\text{safe for typical FPGA GPIO})$$

4.2.3 Component Sourcing and Part Numbers

Table 4: Component Purchasing Information

Component	Part Number	Reference
IR LED (1A, 940nm)	LITEON Infrared (IR) LEDs	[9]
MOSFET (Logic Level)	IRLZ44NPbF (Infineon)	[10]
Resistor 3.3Ω, 5W	280-CR5-3.3-RC (Xicon)	[11]
Resistor 220Ω, 1/4W	CF14JT220R (Stackpole)	[12]
Resistor 10kΩ, 1/4W	CF14JT10K0 (Stackpole)	[13]
Capacitor 100μF, 16V	UPW1C101MDD (Nichicon)	[14]
Capacitor 0.1μF, 50V	K104K15X7RF5TL2 (Vishay)	[15]

4.2.4 Connection Instructions

Follow these connection steps to assemble the circuit:

1. **Power Supply:** Connect +5V to LED anode, C1 (+), C2 (+)
2. **Ground:** Connect all grounds together (power supply (-), C1 (-), C2 (-), R3 bottom, R1 bottom, FPGA GND)
3. **LED:** Anode to +5V, Cathode to MOSFET Drain
4. **MOSFET:** Drain to LED cathode, Source to R3 top, Gate to R2 and R1 junction
5. **Resistors:** R2 (220Ω) from Gate to FPGA pin, R1 (10kΩ) from Gate to GND
6. **FPGA:** Output pin to R2, GND pin to common ground

4.2.5 Operating Principle

When the FPGA outputs a 38kHz square wave (for NEC protocol bursts):

- **HIGH (3.3V):** The MOSFET turns ON, allowing current to flow from +5V through the LED, through the MOSFET, through R3 to GND. The LED emits IR light.
- **LOW (0V):** The MOSFET turns OFF, current stops flowing, the LED is dark.
- The 10kΩ pull-down resistor ensures the MOSFET stays OFF during FPGA power-up or reset.

- The 220Ω gate resistor protects the FPGA output pin from current spikes.
- The 3.3Ω resistor limits the maximum current to approximately 1A.
- Capacitors C1 and C2 provide local energy storage for clean switching at 38kHz.

4.2.6 Thermal Management of the Circuit

Operating at a current of 1 A results in a power dissipation of approximately 1.5 W in the IR LED. This level of power dissipation is substantial and can cause thermal damage to the LED junction if not properly managed. To ensure reliable operation and prevent thermal runaway, it is essential to thermally couple the IR LED to a heatsink with adequate thermal resistance. The heatsink increases the effective surface area for convection, significantly reducing the junction temperature and extending the component's lifespan.

To optimize heat transfer, a suitable thermal interface material (TIM), such as thermal paste, should be applied between the LED package and the heatsink base. For this application, a heatsink with a thermal resistance lower than 20 °C/W under natural convection is recommended.

A suitable candidate is the Aavid 573300D00000G [16] extruded aluminum heatsink, which provides a thermal resistance of approximately 16 °C/W in natural convection.

Figure 7: Aavid 573300D00000G

4.2.7 Important Notes and Warnings

- **Heatsink Required:** The IR LED MUST be mounted on a heatsink. At 1A, it will overheat and fail within seconds without proper thermal management.
- **Resistor Power Rating:** The 3.3Ω resistor dissipates 3.3W of heat. A 5W rated resistor is mandatory. Standard 1/4W resistors will catch fire.
- **Common Ground:** The FPGA ground MUST be connected to the power supply ground. Without this, the circuit will not function.
- **Logic-Level MOSFET:** The IRLZ44N is a logic-level MOSFET that turns fully on with 3.3V-5V gate drive. Standard MOSFETs (e.g., IRF510) require 10V and will not work properly.

- **Capacitor Placement:** Place C1 and C2 as close as possible to the LED and MOSFET to minimize inductance and ensure clean switching.

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